



SECJ2203: Software Engineering

System Documentation (SD)

KADA Cooperation System

Version 1.0

Date

Faculty of Computing

Prepared by: KodeLab

Revision Page

a. Overview

This current version of the documents outlines the requirements for the system, focusing on the earlier development of the system. It included the following sections :

1. Introduction includes the purpose, scope, definitions, acronyms and abbreviations, the references and also the overview of the system and the project.
2. Specific requirements provide the persona, system features, user story, along with the non-functional requirements and also the system's design constraint.

This version of the document reflects the state of the system during the Sprint 0 of the development and can act as the references for stakeholders of KADA Cooperative and the future developers team to understand the developed function and its functionality of the system.

b. Target Audience

1	Documentation writer
2	Project manager
3	Board of director
4	Applicant and member
5	Administrator

c. Project Team Members

Member Name	Role	Task	Status
Lubna Al Haani	Leader	1.5.0 Overview	Complete
		2.1.2 Persona 2	Complete
		2.2.3 State diagram	Complete
		2.4.7 User story 2	Complete
Nabil Aflah Boo	Member	Section introduction	Complete
		1.2.0 Scope	Complete
		2.1.4 Persona 4	Complete
		2.4.1 User story 4	Complete

Nur Firzana	Member	2.1.3 Persona 3	Complete
		2.2.2 Class diagram	Complete
		2.3.0 Launch phase	Complete
		2.4.3 User story 3	Complete
Nurul Adriana	Member	1.1.0 Purpose	Complete
		2.2.1 Use case diagram	Complete
		2.6.0 Design constraint	Complete
Pravinraj	Member	Target audience	Complete
		2.1.1 Persona 1	Complete
		2.4.4 User story 1	Complete
		2.5.0 Functional and Non-functional requirement	Complete

d. **Version Control History**

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Team leader (Lubna Al Haani) Member (Nabil Aflah Boo, Nur Firzana, Nurul Adriana, Pravinraj)	Completed Section 1.0 and 2.0	01/12/2024

Note:

This System Documentation (SD) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998), Software Design Descriptions (SDD) (IEEE Std. 1016-1998 1), and Software Test Documentation (IEEE Std. 829-2008) that are simplified and customized to meet the need of SECJ2203 course at School of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly.

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1. Introduction

The KADA Cooperative System is a website platform designed to reduce the manual work processes of the cooperative. This system provides several functionalities such as managing members, loan applications, and financial reporting. This system helps to boost the cooperative efficiency by enabling the real-time access and automated process of applications.

1.1 Purpose

This Software Requirements Specification (SRS) document presents the KADA Cooperative System, a web-based product intended to enhance the operation of cooperatives through digitization of membership and loan applications. This document outlines the comprehensive scope of development for the system and serves as a guide for evaluating it as a reference for key stakeholder groups, including project managers, developers, testers, and consultants.

Technical Objectives

- a) From this definition, derive the functional and non-functional requirements of the system consistent with those demanded by users and project goals.
- b) Leave a space for iterative development and delivery of features within the Agile framework with clarifications on how this should be implemented for flexibility and continuous improvement.
- c) This acts as a basis for checking how the system complies with legal, security, and quality requirements.

User Benefits

- a) Replace the inefficient, manual processes with a safe, user-friendly online alternative in order to facilitate the collaboration outcome.
- b) Offer an improved accessible interface and real-time updates at any time, to members, administrators, and the Board of Directors.
- c) Save trees and go green by reducing paper consumption.

This SRS shall be meant for internal development teams and external system consultants, who will read and evaluate it as an iteration step in the project. This document is really important for communication between all involved parties and their successful collaboration.

Intended Audience and Reading Suggestions

- 1. Project Managers: Use the Overview, Scope, and Functional Requirements sections to plan timelines and align deliverables with user needs.
- 2. Documentation Writers: Use the entire document to craft user guides for all user groups.

3. Applicants and Members: Refer to the Overview, User Role Definitions, and Functional Requirements sections for membership, loans, and account details.
4. Administrators: Refer to the Administrative Functions section for managing accounts, communications, and reports.
5. Board Members: Refer to the Administrative Functions section for a description of system reports and oversight tools.

1.2 Scope

- a) The software product is called KADA Cooperative System.
- b) The product should also be able to send notifications via email and SMS to the members to inform them their applications status. Some of the functions of the products are apply for membership, apply for loan, view financial statements, manage member and member statements, manage BOD report, review application, and view financial report.
- c) This software product will be used by four target audiences which are applicants, cooperative members, administrators, and board of directors. Each audience will be directed to each different portal where only the specific audience can access the portal. The product helps to minimize the manual work processes and enhance user experience through the role-specific interface. It's to achieve a better understanding of the system architectures and its functions, also to serve as potential enhancement by ensuring adaptability evolves the cooperative's needs.

1.3 Definitions, Acronyms and Abbreviation

Table 1.1 : Definitions, Acronyms and Abbreviation

Terms	Definitions
KADA	Kemubu Agricultural Development Authority
BOD	Board of Director
SD	System Documentation
Admin	Administrator

1.4 References

- [1] Dennis, A., Wixom, B. H., & Tegarden, D. (2015). *Systems Analysis and Design: An Object-Oriented Approach with UML* (A. Dennis, B. H. Wixom, & D. Tegarden, Eds.). Wiley.

1.5 Overview

Following the introduction, the SD document will delve into the specific system requirements. It begins by presenting the personas of the users and the system features, which include detailed use cases, class diagrams, and state diagrams. Additionally, the requirements also included user stories for each use case, along with the functional and non-functional requirements, and design constraints.

The SD document is organized into sections to provide a comprehensive and organized view of the system's requirements aspects. After the introduction, the specific requirements section are divided into subsections and organized as following :

1. Persona

This subsection describes all the system's user types or actors including their user story with the system.

2. System Features

This subsection provides graphical representation of the system: the use case diagram, class diagram and state diagram.

3. Launch Phase

This is where the product backlog and the sprint phase for the system is provided.

4. User Story Details

Each user story is provided with user story description along with supporting graphical representation which is the activity diagrams and sequence diagrams.

5. Performance and Other Requirements

This subsection provided the functional and also the non-functional requirements such as system performance and attributes.

6. Design Constraints

This subsection highlights the constraints that are imposed by the organisation for the systems to adhere such as hardware and security constraints.

2. Specific Requirements

This section of SD will show the details of all software requirements of the KADA Cooperative System. These requirements are specific to guide the designers in creating a system that satisfies users' requirements while enabling developers to implement such features. Besides, it allows the testers to validate whether the system matches its intended purpose. Each of the system requirements will include their inputs, outputs and function to the system. The users are given their specific roles and responsibilities within the system to ensure the system is streamline with their goals effectively. This section should be able to provide a better understanding of the intended audience and form the basis for the design and development of the system.

2.1 Persona

Personas are fictional representations of key users that help to understand their needs, goals and behaviors. It guides the design and development of the system to ensure it meets user expectations effectively.

2.2.1 Persona 1 (Applicant)

User Need

- The applicant need a way to apply for KADA membership

User Stories

1. As an applicant, I want to apply for KADA membership to become a member in KADA and use its loan benefits.

Table 2.1 : An applicant user story 1

An applicant apply for membership
The system must be open for input. Complete the registration form. Fill in the required information and submit it. The system will check if the submitted information is complete and accurate. Then, the application is assessed by the competent authority. If the application is approved as a member, the system generates a member ID. An email is dispatched to the applicant containing login details. If rejected, the system records the grounds for refusal. The email provides the reasons for denial.

2.1.2 Persona 2 (Member)

User Need

- The member need a way to apply for loan, view loan application status and view financial statements

User Stories

1. As a member, I want to apply for a loan so that I can exercise my rights as a member of the cooperative.
2. As a member, I want to view the loan application status so that I can know if my application is successful or not.
3. As a member, I want to be able to view my financial statements in the system so that I know my current financial status.

Table 2.2 : A member user story 1

A member apply for a loan
The system must be open for input. Click on the loan application. Click on the new application. Fill the form with the details needed. If you have completed filling in all the required details on the form, click submit. The system will notify you that the application is successfully accepted into the system reviewing. If you do not fill in some of the required details, click submit. The system will alert you to fill the empty required fields. Then, click submit again after filling in all details. In all cases, the system requires you to login into the system to be able to apply for a loan.

Table 2.3 : A member user story 2

A member view the application status
Click on the loan application. Click on status. If the member has applied for a loan, the system will display the list of loan applications with its status. If the member has not yet applied for a loan, the system will not display any loan application. The system will give you a note that you have not yet applied for a loan. In all cases, the system requires you to login into the system to be able to view financial statements.

Table 2.4 : A member user story 3

A member view the financial statements
The system must be open for input. Click on the statements and choose a month.
If the chosen month's statement is available, the system will display the statement.
If the chosen month's statement is not available, the system will notify you that the statement for that month is not available yet.
In all cases, the system requires you to login into the system to be able to view financial statements.

2.1.3 Persona 3 (Admin)

User Need

- KADA Admin needs a way to manage KADA profile and dashboard, membership applications, loan requests and notify applicants' status and edit reports.

User Stories

1. As a KADA Admin, I want to manage the KADA profile and dashboard so that I can maintain accurate and updated records.
2. As a KADA Admin, I want to approve or disapprove membership applications so that only the eligible members can access the cooperative's services.
3. As a KADA Admin, I want to access and update the status of loan applications so that I can ensure the cooperative's resources are reliable.
4. As a KADA Admin, I want to notify the applicants via email whether their application is approved or disapproved so that they can be alerted with their status application.
5. As a KADA Admin, I want to edit the report so that BOD can review the report easily.

Table 2.5 : An admin user story 1

An Admin manage the KADA profile and dashboard
The system must be open for input. Click on the 'Profil Koperasi' tab. If you want to edit and update the latest information of KADA Cooperative, you can select the pencil icon, the system will give access to edit the information details. Then, you can click the save button. If you want to update the application's statistics, you can click the 'Statistik Permohonan Pinjaman' tab, then click on the pencil icon to edit the updated statistics based on the applications. If you want to update the application's statement, you can click the 'Penyata Ahli' tab, then click on the pencil icon to edit the updated statement and save button. In all cases, the system requires you to login into the system to be able to manage the KADA profile and dashboard.

Table 2.6 : An admin user story 2

An Admin manage Membership Applications
The system must be open for input. Click on the 'Permohonan Ahli' tab to view the list of membership applications. If you want to search the applicants by their name, click on the search engine, then can search the name of the applicants, the system will display the detailed information of the applicants. If the name won't appear, the system will show that the applicant's name is not available. If you click on the applicant's name according to the list, the system will display the detailed information of the applicant. If you want to approve the applications, click yes on the checkbox, the system will update the applicants' status to approved, meaning that they are eligible to be a member of KADA Cooperative. If you want to reject the applications, do not click on the checkbox, the system will update the applicants' status to rejected, meaning that they are not eligible to be a member. In all cases, the system needs you to stay logged in to be able to manage membership applications.

Table 2.7 : An admin user story 3

An Admin process Loan Applications
The system must be open for input. Click on the 'Permohonan Pembiayaan' tab to view the list of loan applications. If you want to search the applicants by their name, click on the search engine, then can search the name of the applicants, the system will display the detailed information of the applicants. If the name won't appear, the system will show that the applicant's name is not available. If you click on the applicant's name according to the list, the system will display the detailed information of the applicant. In all cases, the system needs you to stay logged in to be able to process the loan applications.

Table 2.8 : An admin user story 4

An Admin notify Applicants' Status
The system must be open for input. Click on application status. If you approve the applications, click the approve button, the system will send an email to applicants that their status has been approved. If you want to reject the applications, click on the reject button, the system will send an email to the applicants that their status has been rejected. In all cases, the system needs you to stay logged in to be able to notify applicants' status.

Table 2.9 : An admin user story 5

An Admin edit the Report
The system must be open for input. Click on the 'Laporan Keseluruhan' tab. If you want to edit and update the latest information of a report, you can select the pencil icon, the system will give access to edit the information details. Then, you can click the save button. In all cases, the system needs you to stay logged in to be able to edit the Report.

2.1.4 Persona 4 (Board of Director)

User Need

- The Board of Director needs a way to review members' applications, loan applications, and view financial summary.

User Stories

1. As a Board of Director, I want to view a list of all applicants including their profiles and details so that I can monitor all the members' activity effectively.
2. As a Board of Director, I want to review all the loan applications so that I can make a decision on the applicant's loan status during the annual meeting.
3. As a Board of Director, I want to view the details of the financial summary so that I can oversee the financial conditions and minimize financial risks.

Table 2.10 : A BOD user story 1

A BOD review members' application
The system must be open for any input. Click on the 'Permohonan Ahli' section to view the members application details. If you want to search for a member's name, click on the search engine and type the particular name. The system will then display the details of the members application. If the name won't appear after the search, the system will show that the member's not available. If you don't want to search the member's name, scroll down the list of tables to see all the members' applications. If you want to view detailed information of the members application, click on the blue words 'Click Here' to see their application form details. In all cases, the system needs you to stay log in to review the members' applications.

Table 2.11 : A BOD user story 2

A BOD review loan applications
The system must be open for any input. Click on the 'Permohonan Pembiayaan' section to view the loan application details. If you want to search for the name of the loan applicant, click on the search engine and type in the name. If the name won't appear after the search, the system will display that the applicants is not available. If you want to view all loan applicants, scroll down the list of tables to see their loan details. If you want to view their loan application details, click on the 'Click Here' to be able to redirect to their forms application detail. In all cases, the system needs you to stay logged in to be able to review the members' loan applications.

Table 2.12 : A BOD user story 3

A BOD view financial summary
The system must be open for any input. Click on the 'Rumusan Kewangan' to view financial statistics. The system will display a profit rate for month, total dividend by year, and statistics for active loan by month for the financial summary dashboard. If you want to see the dividend history for KADA, click on the download icon or scan the qr code to review it. If you want to review the financial report, click on the download icon or scan the qr code to see the overall financial report for the current year. In all cases, the system needs you to stay logged in to be able to view the financial summary.

2.2 System Features

This product is a new website system for KADA Cooperative which is aimed at making a complete change from the hard copy methods of processing membership and loan applications to more automated systems. It tries to help the user eliminate all the inconveniences that are associated with applying for membership and loans, managing user details and even managing reports. In addition, the system incorporates the capabilities of calculating cooperatives' loans and interests for each member, reducing the workload on the people.

The system features are illustrated in Figure 2.1 below. The detailed description of each module and functions is tabulated in Table 2.13.

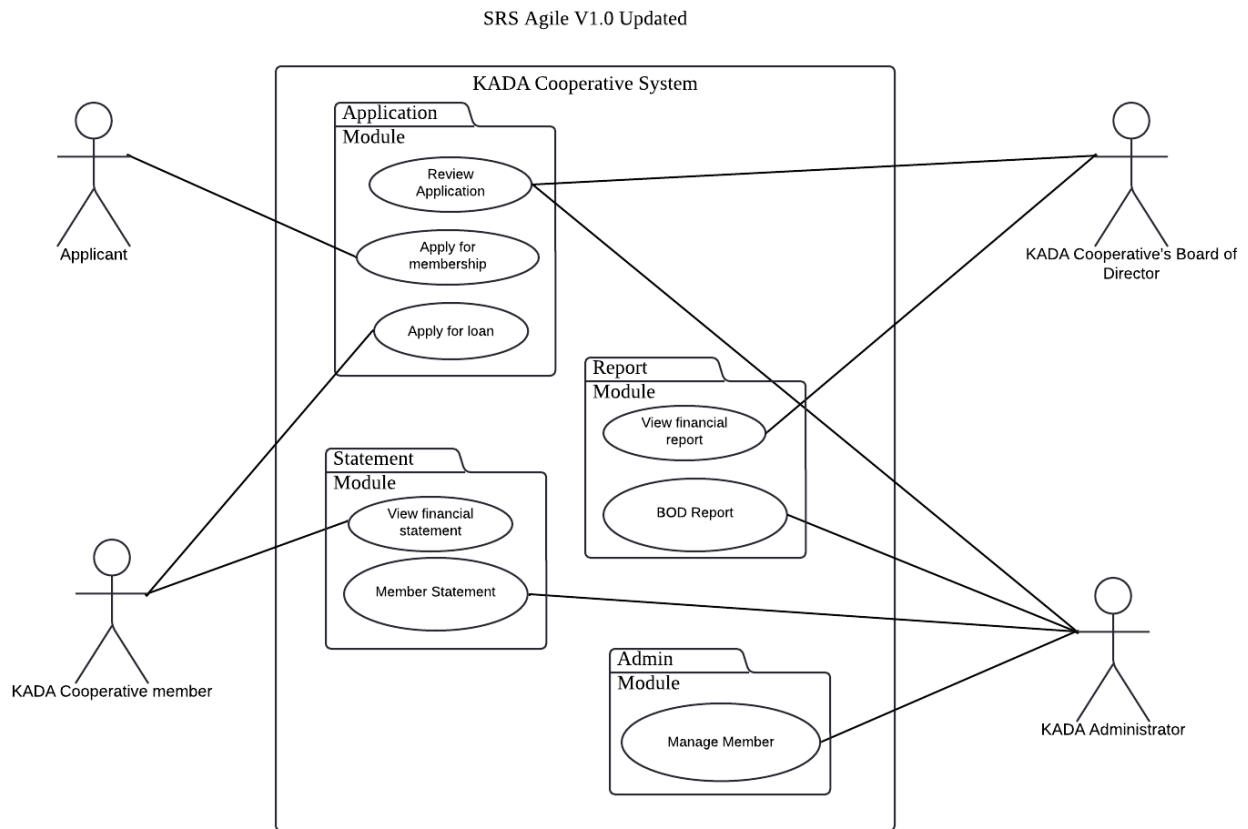


Figure 2.1: Use Case Diagram for KADA Cooperative System

Table 2.13 : Description of Module and Functions for KADA Cooperative System

Use case	Function	Description
UC001	Apply Membership	This use case allows individuals to apply for membership by providing required personal details.
UC002	Apply Loan	This use case allows members to submit a loan application along with necessary documents.
UC003	Review Application	This use case enables administrators to review and either approve or reject membership and loan applications.
UC004	View Financial Statement /Member Statement	This use case allows both members and administrators to view member-specific financial details, including balances and transaction histories.
UC005	View Financial Report /BOD Report	This use case allows administrators or Board of Directors (BOD) members to view comprehensive financial reports for the organization. This use case also provides detailed reports specifically designed for Board of Directors' decision-making purposes.
UC006	System Administration	This use case allows system administrators to manage user roles, permissions, and system settings.

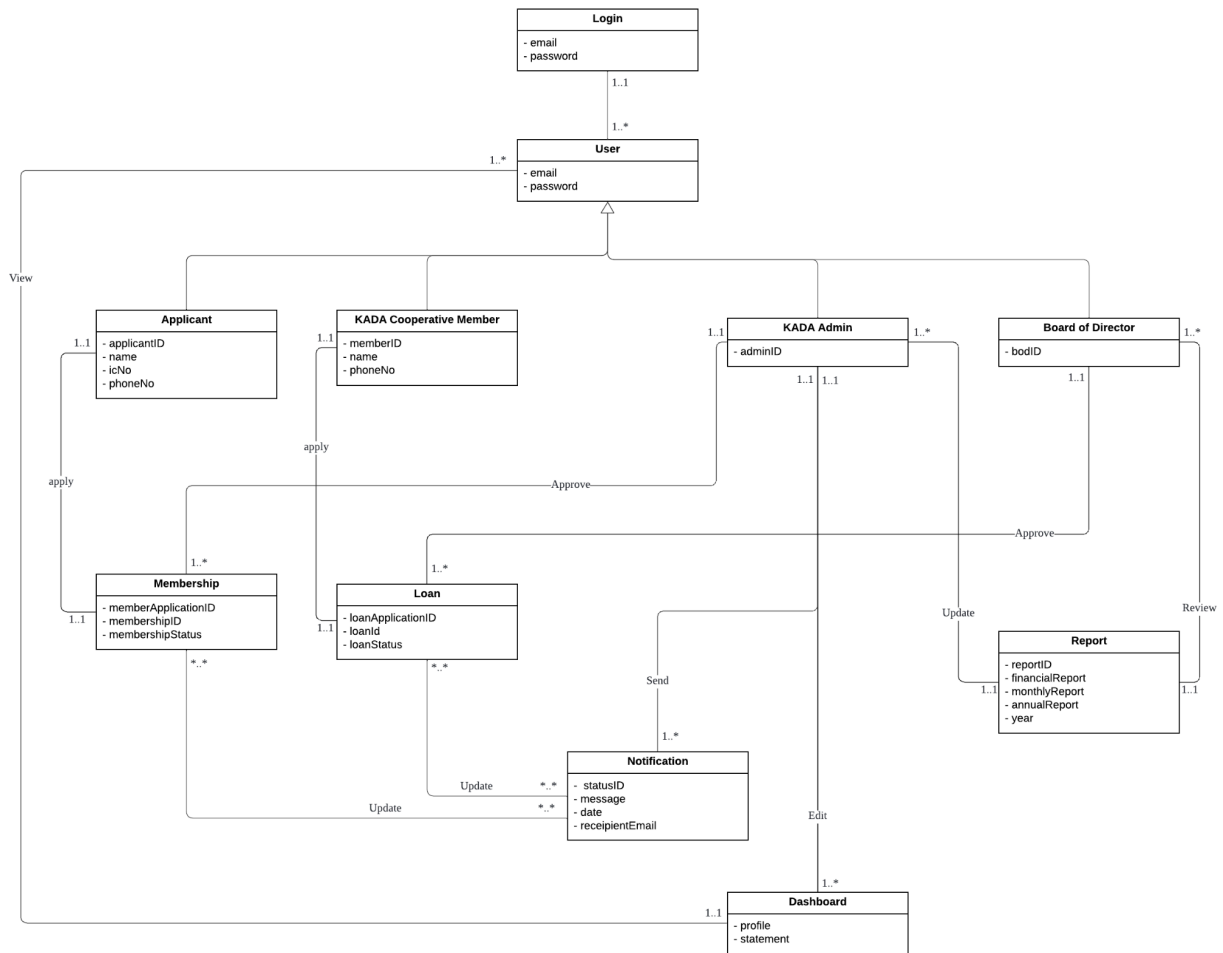


Figure 2.2 : Domain Model for KADA Cooperative System

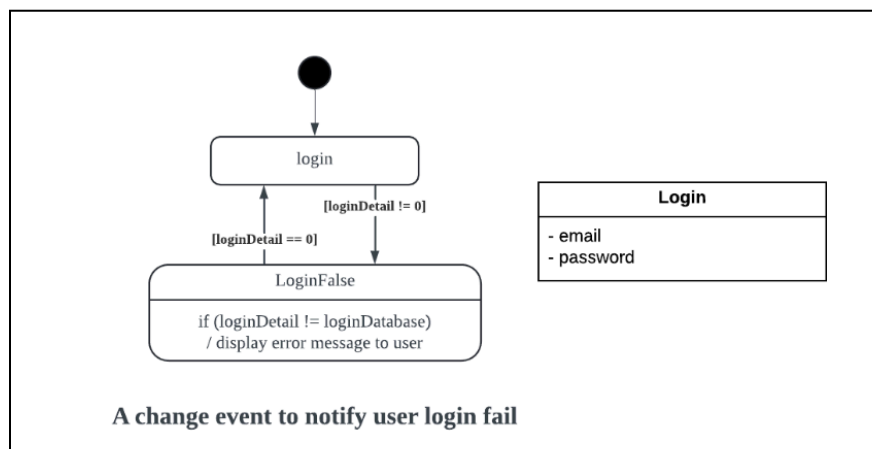


Figure 2.3 : State Machine Diagram for Login

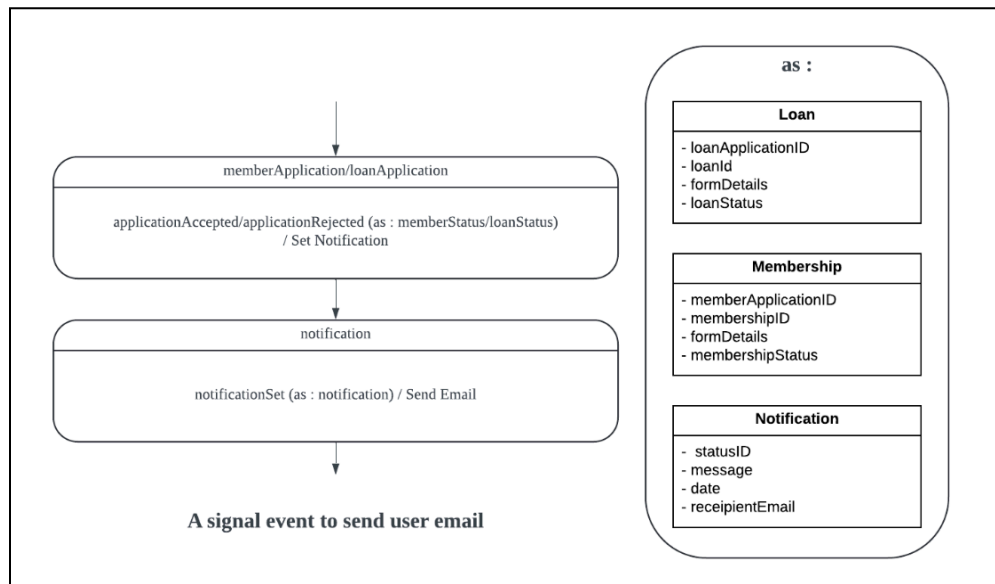


Figure 2.4 : State Machine Diagram for Membership, Loan and notification

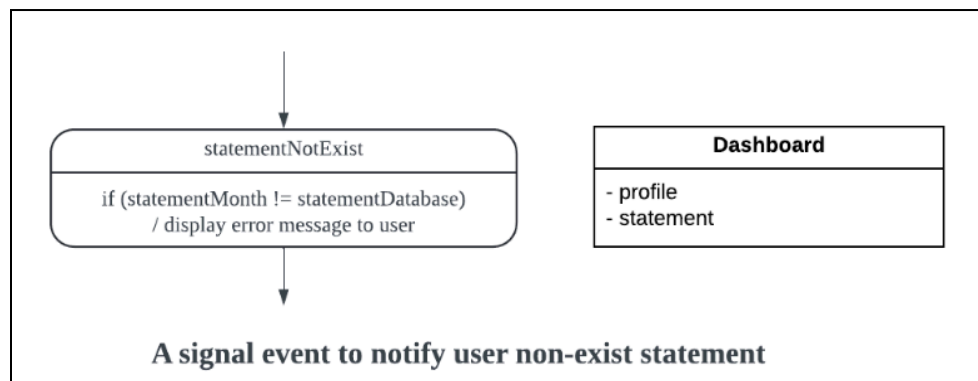


Figure 2.5 : State Machine Diagram for Dashboard

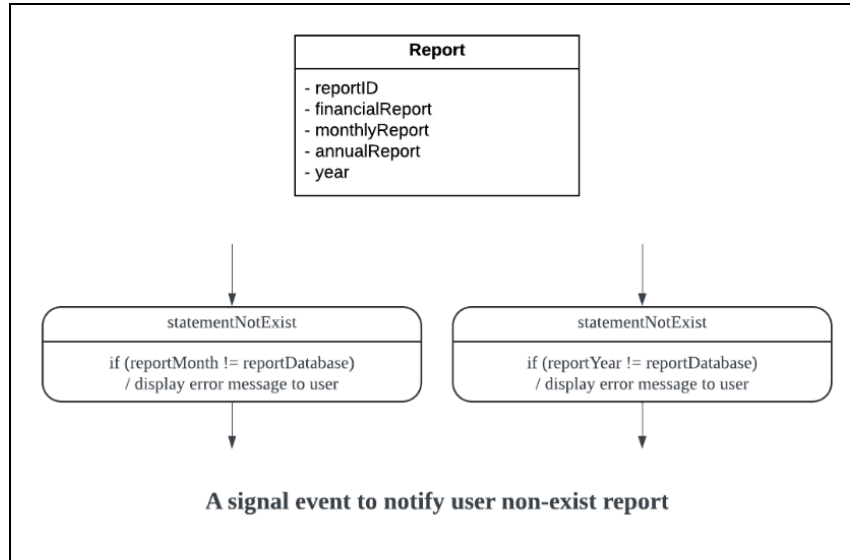


Figure 2.6 : State Machine Diagram for Report

2.3 Launch Phase

The product backlog includes several sprints to develop the key features of the KADA Cooperative System. Table 2.3 lists the sprints, user stories and assigned team members.

Table 2.14 : Launch Phase

Sprint	Team members assigned
Sprint #1 An applicant wants to apply for KADA membership to become a member in KADA and use its loan benefits.	Pravinraj
Sprint #2 A member wants to apply for a loan so that I can exercise my rights as a member of the cooperative.	Haani
Sprint #3 A KADA Admin wants to approve or disapprove membership applications so that only the eligible members can access the cooperative's services.	Firzana
Sprint #4 A KADA Admin wants to notify the applicants via email whether their application is approved or disapproved so that they can be alerted with their status application.	Firzana
Sprint #6 A Board of Director wants to review all the loan applications so that I can make a decision on the applicant's loan status during the annual meeting.	Aflah
Sprint #6 A Board of Director wants to view the details of the financial summary so that I can oversee the financial conditions and minimize financial risks.	Aflah

User Story Details

2.4.1 US001 User Story : Applicant

Table 2.15 : User Story Description for Applicant

User story: Applicant
ID: US001
User Story Description As a applicant I want to apply for KADA membership So that I can become a member of KADA and use its loan benefits.
Flow of Events: 1. The applicant signed in into the system. 2. If the applicant wants to apply for KADA membership: 2.1. The applicant navigates to the registration form section. 2.2. The applicant clicks on "New Application." 2.3. The applicant fills out the registration form with the required details. 3. If all required details are completed: 3.1. The applicant clicks "Submit." 3.2. The system confirms the successful submission of the application. 4. If some required details are missing: 4.1. The applicant clicks "Submit." 4.2. The system alerts the applicant to fill in the missing fields. 4.3. The applicant fills in the required details and clicks "Submit" again.
Alternative flow 1:
If the applicant has not yet applied for KADA membership: 1. The system notifies the applicant that they fill the registration form first and get membership.
Acceptance Criteria - Precondition : The applicant must be signed in to apply for KADA membership - Postcondition : The membership application's information is recorded in the system - Other condition: The applicant's information are validated during the registration process to check completeness and accuracy of data
Exception Flow If the system detect any issues because of invalid data or missing required fields: The applicant prompt to correct the error and resubmit the form

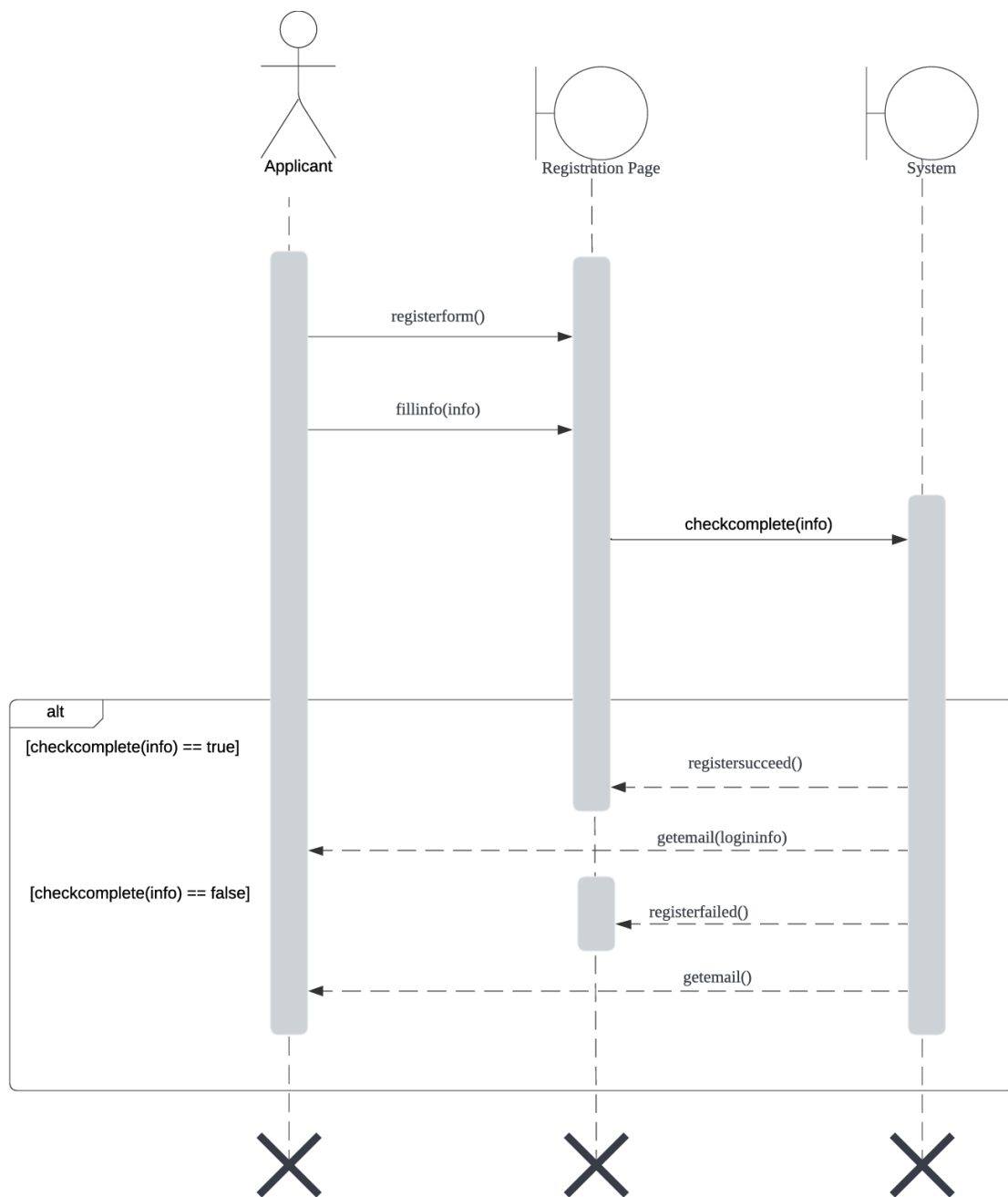


Figure 2.7 : Sequence Diagram for Applicant

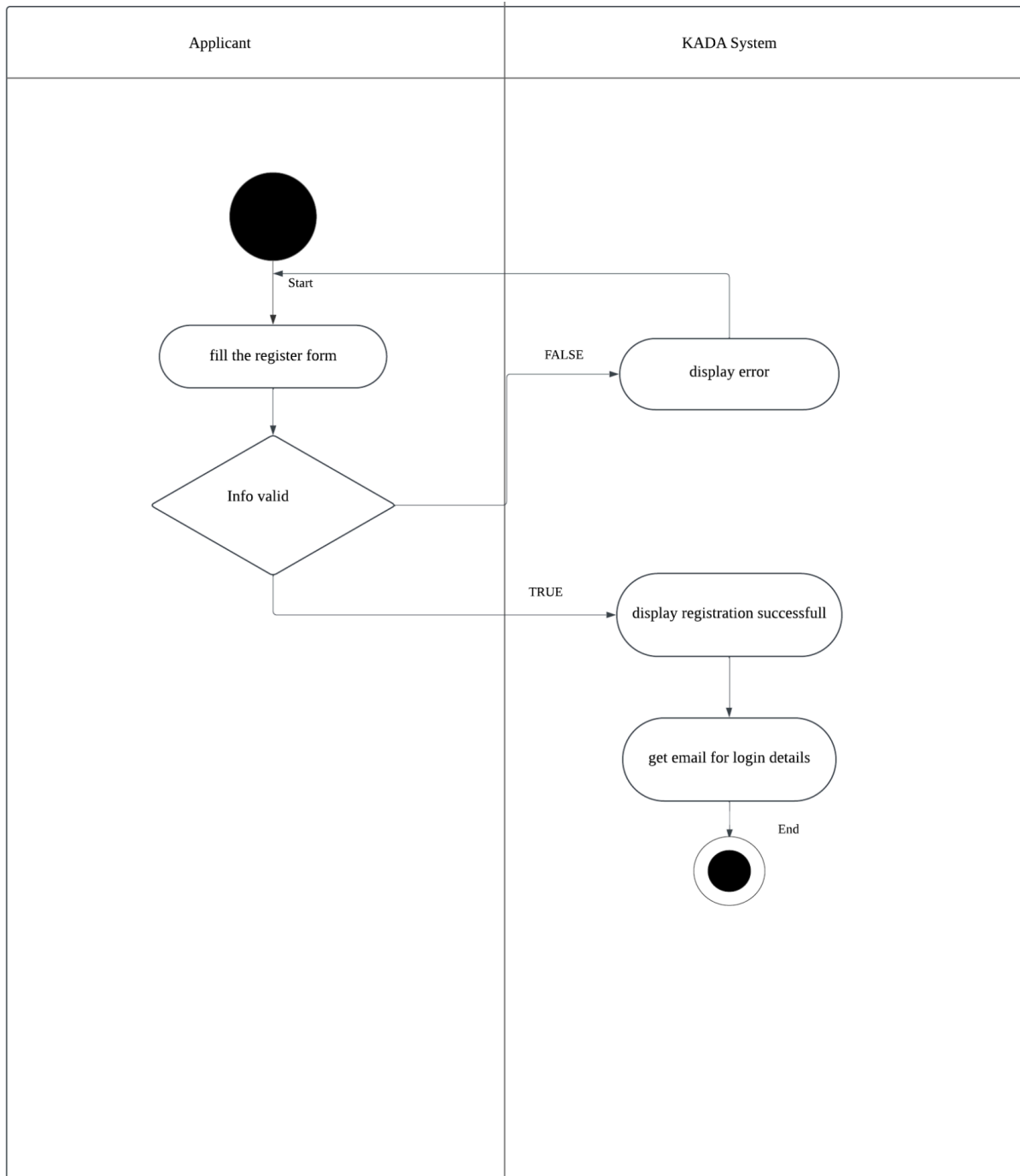


Figure 2.8: Activity Diagram for Applicant

2.4.2 US002 User Story : Member

Table 2.16 : User Story Description for Member

User story: Member
ID: US002
User Story Description 1. As a member, I want to apply for a loan so that I can exercise my rights as a member of the cooperative. 2. As a member, I want to view the loan application status so that I can know if my application is successful or not. 3. As a member, I want to be able to view my financial statements in the system so that I know my current financial status.
Flow of Events: 1. The member logs into the system. 2. If the member wants to apply for a loan. 2.1. The member navigates to the loan application section. 2.2. The system displays available loan options and eligibility criteria. 2.3. The member clicks on a new application. 2.4. The member fills out the loan application form and submits it. 2.5. The system confirms the application submission and provides a reference number. 3. If the member wants to view loan application status. 3.1. The member navigates to the loan application section. 3.2. The system will display available loan options and eligibility criteria. 3.3. The member navigates to the status application. 3.4. The system will display all of the loan applications applied by the member with its status. 3.5. The member can view the loan application status. 4. If the member wants to view the financial statements. 4.1. The member navigates to the financial statements section. 4.2. The system will display the financial statements page. 4.3. The member will choose a month. 4.4. The system will display the chosen month's financial statement. 4.5. The member can view, download or print the financial statements.
Alternative flow 1: • If the member does not yet applied for a loan when view loan application status: 1. The system notifies the member that there is no loan application to display.

Alternative flow 2:

- If the monthly financial statement is not yet available when view financial statement:
 1. The system notifies the member of an unavailable statement for the chosen month.

Acceptance Criteria

- Precondition: The user is an approved member of the cooperative.
- Postcondition: The loan application is recorded in the system, and a reference number is generated.
- Other conditions: The member's details are validated during the process.

Exception flow:

If the system encounters a technical error while processing the loan application, the user is notified of the error, and the application is saved in a draft state for later completion.

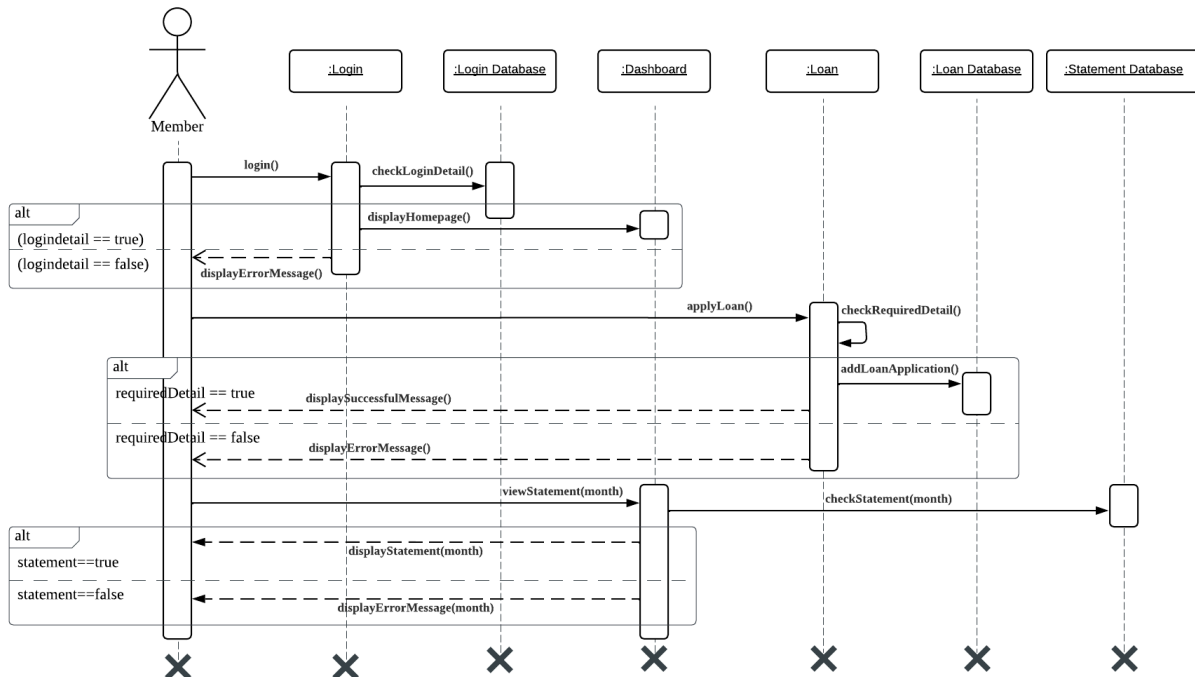


Figure 2.9: Sequence Diagram for Member

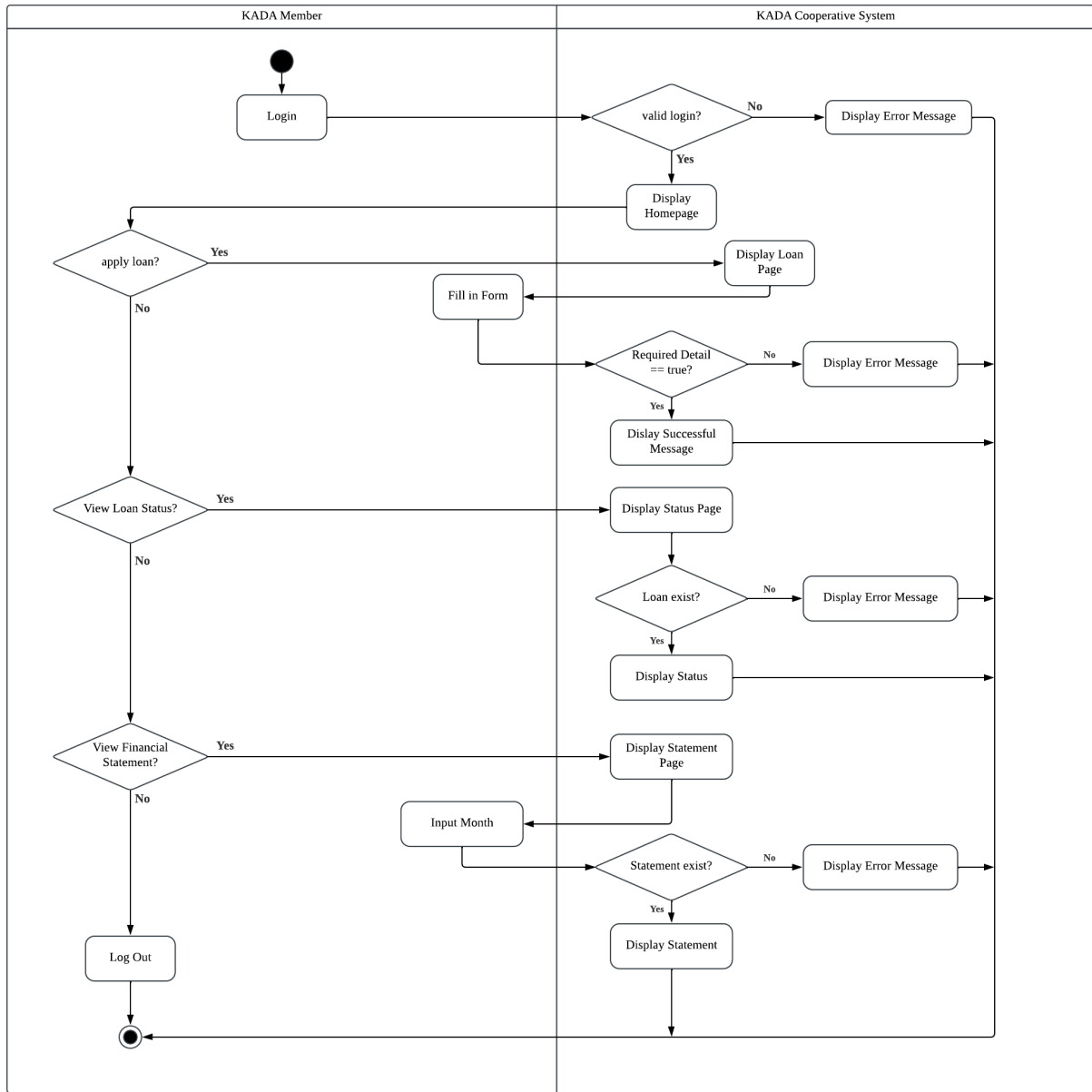


Figure 2.10 : Activity Diagram for Member

2.4.3 US003 User Story : Admin

Table 2.17 : User Story Description for Admin

User story: Admin
ID: US003
User Story Description 1. As a KADA Admin, I want to manage the KADA profile and dashboard so that I can maintain accurate and updated records. 2. As a KADA Admin, I want to approve or disapprove membership applications so that only the eligible members can access the cooperative's services. 3. As a KADA Admin, I want to access and update the status of loan applications so that I can ensure the cooperative's resources are reliable. 4. As a KADA Admin, I want to notify the applicants via email whether their application is approved or disapproved so that they can be alerted with their status application. 5. As a KADA Admin, I want to edit the report so that BOD can review the updated report easily.
Flow of events: 1. An admin logs into the system. 2. If an Admin wants to manage the KADA profile and dashboard. 2.1. The system will display an admin dashboard with cooperation profile, statistics of loan application and member statements. 2.2. An Admin navigates to a cooperation profile or statistics of loan application or member statements. 2.3. The system will display the editable page with a pencil icon. 2.4. An Admin clicks on a pencil icon and can update the latest information. 3. If an Admin wants to manage membership applications. 3.1. An Admin navigates to the membership application tab. 3.2. The system will display the list of membership applicants. 3.3. An Admin clicks on the name of the applicant. 3.4. The system will show the detailed information of the applicant. 3.5. An Admin clicks the checkbox to approve the applications. 3.6. The system will update the status of the applications. 4. If an Admin wants to process loan applications. 4.1. An Admin navigates to the loan application tab. 4.2. The system will display the list of loan applicants. 4.3. An Admin clicks on the name of the applicant. 4.4. The system will show the detailed information of the applicant. 4.5. An Admin updates the application status. 5. If an Admin wants to notify applicants' status. 5.1. An Admin navigates to the application status tab. 5.2. The system will display the list of applications. 5.3. An admin clicks membership applications or loan applications option. 5.4. The system will display the list of choosed applications. 5.5. An Admin clicks on the approve or reject button. 5.6. The system will send the notifications via email to the applicants. 6. If an Admin wants to update the report. 6.1. An Admin navigates to the overall report tab. 6.2. The system will display the data of the report. 6.3. An Admin clicks on the report name.

- 6.4. The system will show an editable report.
- 6.5. An Admin clicks on the pencil icon and save button after editing.
- 6.6. The system will update the latest version of the report.

Alternative flow *n*:

- If the members is not recorded:
 1. The system displays an error message “Not Available”.
- If Admin rejects membership applications:
 1. The system notify members that their application is being rejected.

Acceptance Criteria

- Postcondition : The system will auto-generate the graph of application statistics.
- Precondition : User logs must be an authorized KADA Admin with appropriate system access.

Exception flow:

1. If the authorization is incorrect, the system will show an error message “Invalid email ID or Password”.
2. If the session times out, the system will redirect the Admin to the login page.

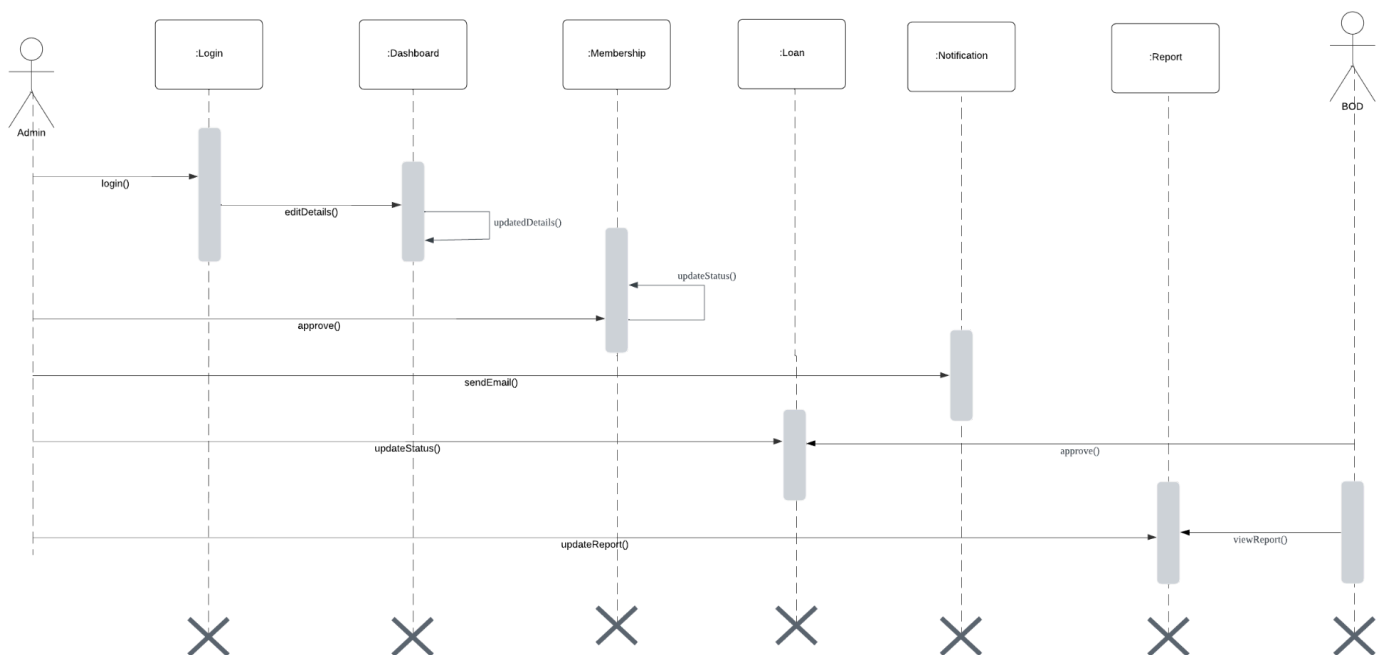


Figure 2.11 : Sequence Diagram for Admin

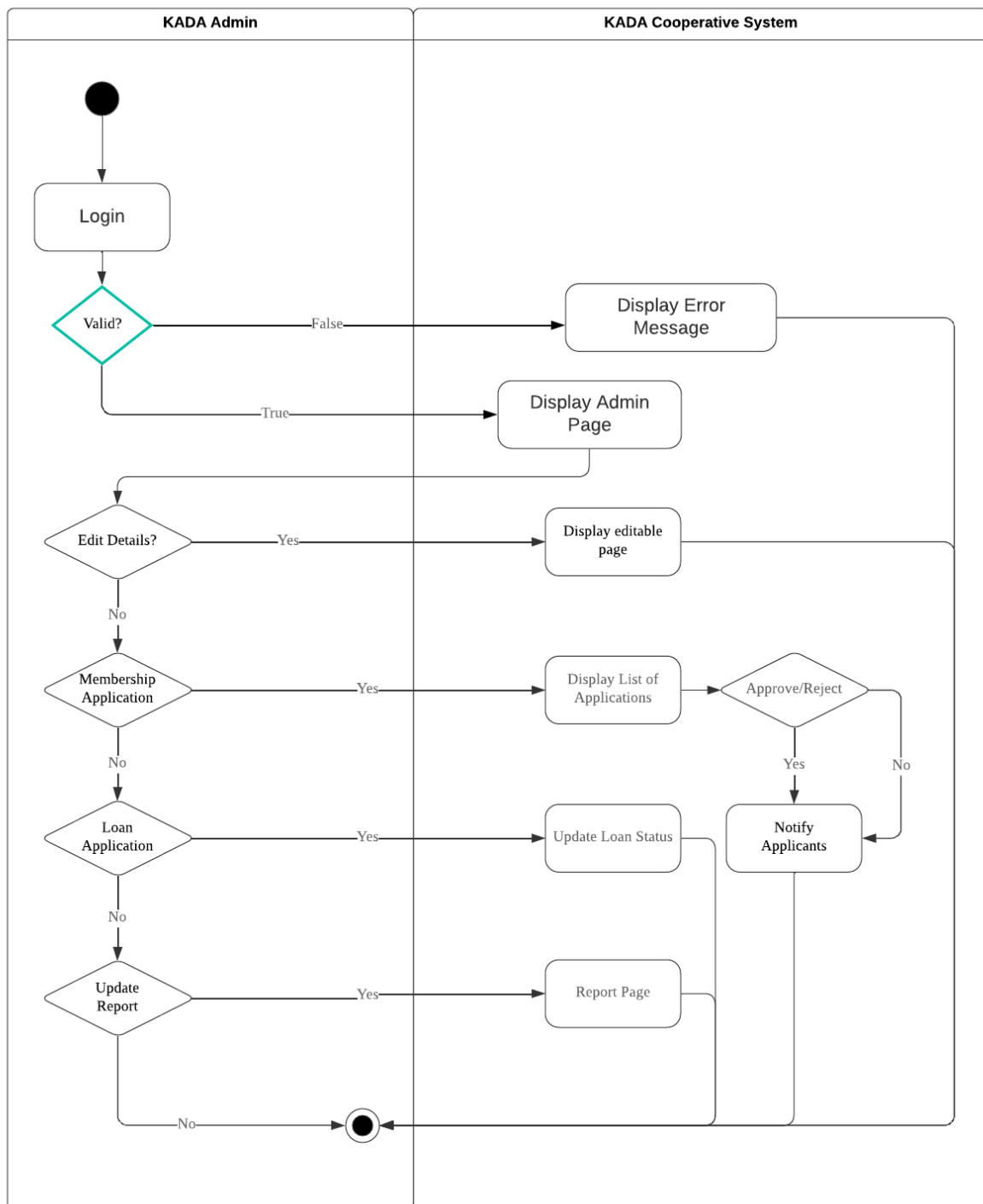


Figure 2.12 : Activity Diagram for Admin

2.4.4 US004 User Story : Board of Director

Table 2.18 : User Story Description for BOD

User story: Bod
ID: US004
User Story Description As a KADA Cooperative's BOD. I want to review loan applications and financial reports. So that I can manage the cooperative's applications status and reduce financial risk.
Flow of events: 1. The BOD logs into the system. 2. The system shows a dashboard with options for "loan application", and "financial summary". 3. The BOD navigates to the "loan application" section. 4. The system displays a list of members who apply for a loan including the types, amount for the loan and applicant details. 5. The BOD navigates to the "financial summary" section. 6. The system displays financial statistics and qr code for the selected year financial report. 7. The BOD scans the qr code or downloads the financial report accordingly.
Alternative flow 1: If the members is not recorded: • The system displays an error message "No such members".
Acceptance Criteria • Postcondition: The BOD discusses the applicant's loan status and financial condition to update to the admin during the annual meeting. • Precondition: The applicants must be recorded into the system for BOD to review.
Exception flow: 3. If the log in credential is incorrect, the system will show an error message "Invalid email ID or Password". 4. If the session times out, the system will redirect the BOD to the login page.

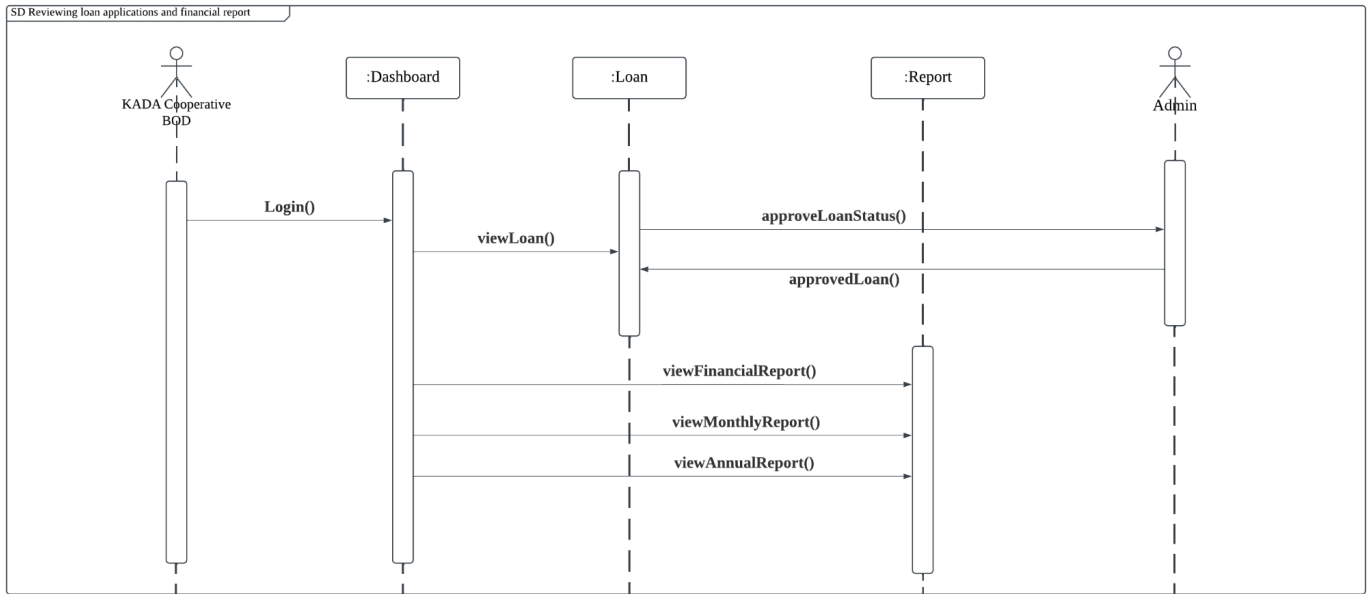


Figure 2.13 : Sequence Diagram for BOD

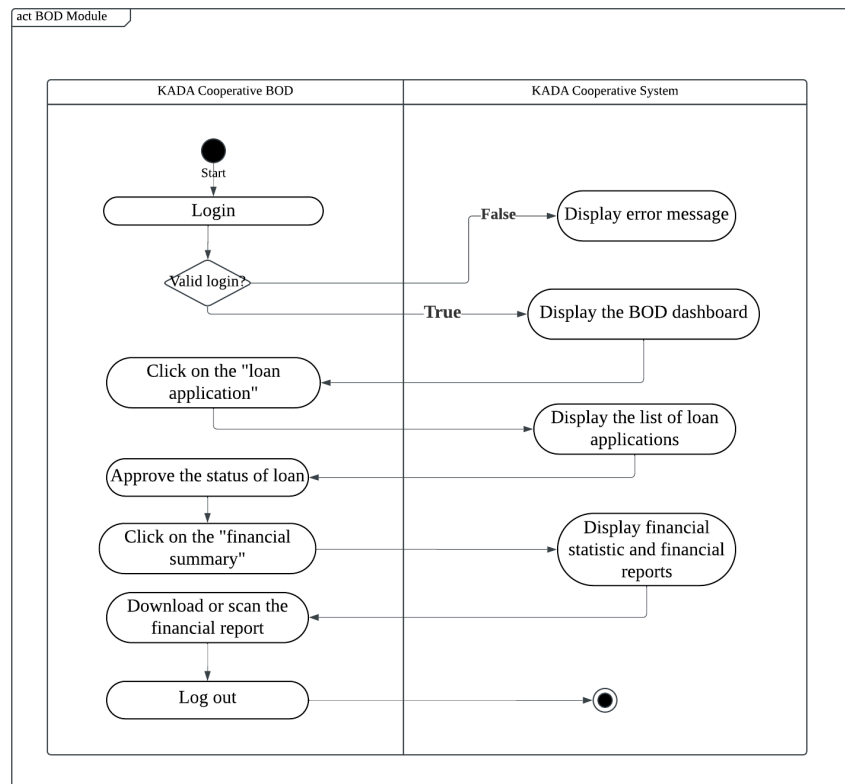


Figure 2.14 : Activity Diagram for BOD

2.5 Performance and Other Requirements

1. Software System Attributes :

Usability: The system shall provide a user-friendly interface for all user roles.

Reliability: Ensure the system features-dashboards, loan applications, and reporting-operate consistently.

Maintainability: Policy parameters such as interest rates or repayment terms shall be updated within 10 minutes without any downtime.

Portability: The system shall be compatible with desktops, tablets, and smartphones and provide a consistent interface.

Compatibility: The system should integrate well with payroll systems for efficient loan repayments.

2. Performance Requirements

Performance requirements define the responsiveness of the system and its capacity to handle data and user requests:

Response Time: User authentication and dashboard loading should not take more than 2 seconds.

Member data and loan retrieval should not take more than 3 seconds.

Throughput: More than 50 transactions per second at peak load.

Capacity: 1,000 concurrent users and data of 100,000 members without any drop in performance.

Availability: 99.9% uptime, with only 8 hours of allowed annual downtime.

Other Requirements

These requirements are about compliance, safety, and environmental considerations:

3. Security: Data security through encryption should be assured via AES-256, along with multi-factor authentication for sensitive operations.

Safety: The system shall prevent unauthorized access via RBAC and shall have detailed audit trails to track all activities.

Legal and Regulatory: Ensure compliance with data protection laws such as Malaysia's PDPA and international standards like ISO/IEC 27001.

Environmental: Optimize resource usage to ensure CPU and memory do not exceed 75% during normal operations.

2.6 Design Constraints

a) *Environmental Limitations:*

- The system should operate efficiently under the common temperature range of the normal office that moves between 15oC and 35oC and relative humidity levels between 30% and 70%.
- Apart from this, the system should perform satisfactorily indoors under normal lighting conditions.

b) *Hardware limitations:*

The system should be deployable on hardware having the following minimal configuration:

- Processor: 2.0 GHz quad-core.
- Memory: 8GB RAM
- Storage: 100 GB of accessible disk space.

c) *Security Regulations:*

- Sensitive user data, such as personal and financial information, should be encrypted using AES-256 when sent and stored.
- Complementary factors are required for member logins and administrator access.
- Administrative functions will be role-based and logged for audit purposes.

d) *Compatibility Constraints:*

- In addition, these systems should work with the latest web browsers like Chrome, Mozilla Firefox, and Microsoft Edge.
- It should be compatible with all newer Windows servers starting from Windows Server 2019, and with Linux-based servers as for instance, Ubuntu 20.04.

e) *Performance constraints:*

- The system has to handle a maximum concurrency of 500 users and has a minor response time of less than 3 seconds for any of its major transactions like loan application.
- While processing loan applications, the batch shall not contain more than 1000 records within a processing window of two minutes.